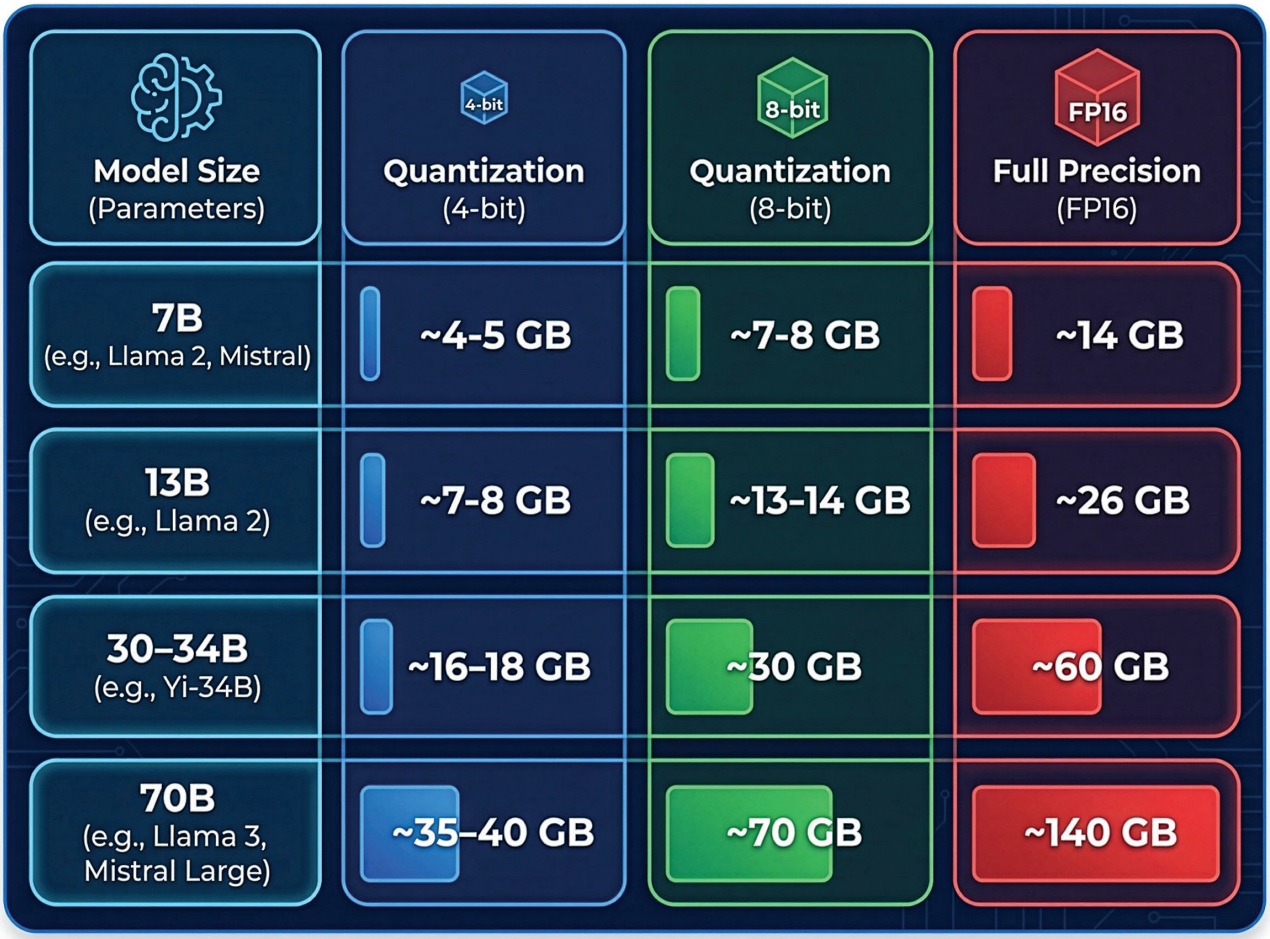


As a rough selection strategy, think in bands rather than exact cut-offs.

Sub-5B models are the first place to look on modest systems. They can be surprisingly capable for rewriting, extraction, simple chat and lightweight coding, especially when highly optimised.

7B–14B-class models are often a useful middle ground for mainstream enthusiasts. They generally offer a noticeable jump in capability while remaining practical in quantised form on many PCs with sufficient memory.

20B–35B-class models can offer stronger reasoning and language quality, but memory pressure rises sharply. Mixture-of-Experts designs complicate the picture because their active parameter count may be much lower than their total size.

70B and beyond should trigger a hardware reality check. Quantisation, large unified-memory Macs, multi-GPU desktops or serious system RAM can make some of these models possible, but "possible" is not the same as pleasant.

OpenAI's gpt-oss-20b shows why model architecture and packaging matter. It contains about 20.9B total parameters but only about 3.6B are active per token. Its MoE weights are natively quantised, producing a 12.8GiB checkpoint that OpenAI says can run on systems with about 16GB of memory. The much larger gpt-oss-120b requires roughly 80GB.

Do not buy a new GPU yet. Chapter 4 will map models to hardware more precisely. For now, use model size as an elimination filter.

Step 4: Choose capability before context-window bragging rights

Model pages love gigantic context numbers. They are easy to market and easy to compare. They are also easy to misuse.

A 128K or 256K context window can be valuable if you routinely analyse large documents, long conversations or codebases. If you mainly ask ten-line questions, it contributes little to everyday usefulness and may increase memory requirements when you actually use a large context.

Meta's Llama 4 Scout is a perfect example of why specifications need interpretation. Meta advertises a context window reaching 10 million tokens, while Scout has 17B active parameters and 109B total parameters. It is natively multimodal and technically impressive, but the total weight footprint means it is not automatically the obvious choice for a mainstream laptop merely because its active count says 17B. Meta itself notes that